CSCI 544: Homework Assignment 1

Sentiment Analysis on Amazon Product Reviews

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1 Dataset Preparation

1.1 Data Source and Loading

For this assignment, I worked with the Amazon Reviews dataset containing customer reviews for Office Products. The dataset was obtained from the Amazon Reviews public dataset repository and loaded using Pandas with tab-separated parsing.

After loading, I retained only the two relevant columns for sentiment analysis:

- `review_body`: The actual text content of customer reviews
- `star_rating`: Numerical rating from 1 to 5 stars

Rows containing missing values were removed, and the star ratings were converted to integer format to ensure consistency throughout the analysis.

1.2 Sample Reviews

Below are three sample reviews from the dataset along with their ratings:

Sample	Rating	Review
1	5 stars	Great product.
2	5 stars	What's to say about this commodity item except, what have we come to in this world. Having the need to buy captured and compressed air.
3	5 stars	Haven't used yet, but I am sure I will like it.

1.3 Rating Distribution

The distribution of ratings across the entire dataset is as follows:

Rating	Number of Reviews
1 star	306,967
2 stars	138,381
3 stars	193,680
4 stars	418,348
5 stars	1,582,704

The dataset shows a strong skew toward positive reviews, with 5-star ratings comprising the majority. This is a common pattern in product review datasets, as satisfied customers tend to leave more reviews.

1.4 Binary Classification Labels

To frame this as a binary sentiment classification problem, I mapped the ratings into three categories:

- **Positive**: Ratings greater than 3 (i.e., 4 and 5 stars) → Label: 1
- **Negative**: Ratings less than or equal to 2 (i.e., 1 and 2 stars) → Label: 0
- **Neutral**: Rating of exactly 3 → Discarded

The resulting class distribution:

Class	Number of Reviews
Positive	2,001,052
Negative	445,348
Neutral	193,680

1.5 Balanced Sampling

Given the significant class imbalance and computational constraints, I randomly sampled 100,000 reviews from each class (positive and negative) to create a balanced dataset of 200,000 reviews. This approach ensures that the classifiers are not biased toward the majority class and allows for meaningful evaluation metrics.

The dataset was then split into 80% training (160,000 reviews) and 20% testing (40,000 reviews) sets using `random_state=42` for reproducibility.

2 Data Cleaning

Raw text data from online reviews contains various noise elements that can interfere with model training. I implemented the following cleaning pipeline:

2.1 Cleaning Steps

1. **Lowercase Conversion**: All text was converted to lowercase to ensure uniformity (e.g., "Great" and "great" are treated identically).
2. **HTML Tag Removal**: Used BeautifulSoup to parse and extract plain text from any HTML markup present in reviews.
3. **URL Removal**: Applied regular expressions to identify and remove web addresses, as URLs do not contribute to sentiment.
4. **Contraction Expansion**: Expanded contractions to their full forms to standardize the text. For example:
   - "won't" → "will not"
   - "can't" → "cannot"
   - "don't" → "do not"
   - "I'm" → "I am"

   I included an extensive dictionary covering common English contractions including informal ones like "gonna" → "going to" and "wanna" → "want to".
5. **Non-Alphabetical Character Removal**: Removed all characters except letters and spaces, eliminating punctuation, numbers, and special characters.
6. **Extra Whitespace Removal**: Collapsed multiple consecutive spaces into single spaces and trimmed leading/trailing whitespace.

2.2 Impact on Review Length

The cleaning process reduced the average review length by approximately 15 characters (4.7% reduction), primarily due to the removal of HTML tags, URLs, and special characters.

Metric	Average Character Length
Before Cleaning	318.0072
After Cleaning	302.9584

3 Preprocessing

After cleaning, I applied natural language preprocessing techniques using the NLTK library to further normalize the text.

3.1 Preprocessing Steps

1. **Stop Word Removal**: Removed common English stop words (e.g., "the", "is", "at", "which") using NLTK's predefined stop word list. These words appear frequently but carry minimal semantic meaning for sentiment analysis.
2. **Lemmatization**: Applied WordNet lemmatizer to reduce words to their base dictionary forms. Unlike stemming, lemmatization considers the context and converts words to meaningful base forms:
   - "running" → "run"
   - "better" → "good"
   - "markers" → "marker"

3.2 Sample Transformations

Below are three examples showing the complete transformation pipeline:

Sample 1:

- **Original**: "This ink did not work on my Epson WF-2630. Printer would not recognize and therefore would not print. Tried various times, rebooted printer etc...The customer service at Amazon were amazing in their understanding and issued me a refund."
- **After Cleaning**: "this ink did not work on my epson wf printer would not recognize and therefore would not print tried various times rebooted printer etc the customer service at amazon were amazing in their understanding and issued me a refund inconvenient to me as a busy buyer as have to order more now"
- **After Preprocessing**: "ink work epson wf printer would recognize therefore would print tried various time rebooted printer etc customer service amazon amazing understanding issued refund inconvenient busy buyer order"

Sample 2:

- **Original**: "These chalk markers have a great medium sized tip and erase well with a wet cloth. So far, I'm loving them and I hope they last throughout the school year."
- **After Cleaning**: "these chalk markers have a great medium sized tip and erase well with a wet cloth so far i am loving them and i hope they last throughout the school year"
- **After Preprocessing**: "chalk marker great medium sized tip erase well wet cloth far loving hope last throughout school year"

Sample 3:

- **Original**: "The ink runs out too fast. Not very good product."
- **After Cleaning**: "the ink runs out too fast not very good product"
- **After Preprocessing**: "ink run fast good product"

3.3 Impact on Review Length

Preprocessing reduced the average length by approximately 115 characters (38% reduction), primarily due to stop word removal. This significant reduction removes noise while retaining the content-bearing words essential

Metric	Average Character Length
Before Preprocessing	302.9584
After Preprocessing	187.4218

for sentiment classification.

4 Feature Extraction

For feature extraction, I used bigram (two consecutive words) representation rather than unigrams. Bigrams capture local word context and can be more informative for sentiment analysis. For instance, "not good" as a bigram carries different sentiment than "not" and "good" separately.

I used scikit-learn's `CountVectorizer` with `ngram_range=(2, 2)` to create a bag-of-bigrams representation. Each review is transformed into a sparse vector where each dimension corresponds to a unique bigram, and the value represents the count of that bigram in the review.

5 Perceptron

The Perceptron is a linear binary classifier that learns by iteratively adjusting weights based on misclassified examples. It serves as a simple baseline for our classification task.

Results

Metric	Training Set	Testing Set
Accuracy	0.9894	0.8572
Precision	0.9833	0.8409
Recall	0.9958	0.8829
F1-Score	0.9895	0.8614

Analysis

The Perceptron achieves near-perfect training performance (98.94% accuracy) but shows a noticeable drop on the test set (85.72% accuracy). This gap of approximately 13 percentage points indicates overfitting to the training data. The high recall (88.29%) on the test set suggests the model is effective at identifying positive reviews, though precision is somewhat lower (84.09%), indicating some false positives.

6 Support Vector Machine (SVM)

I used Linear SVM (LinearSVC) for this task. SVMs find the optimal hyperplane that maximizes the margin between classes, making them well-suited for high-dimensional text classification problems.

Results

Metric	Training Set	Testing Set
Accuracy	0.9917	0.8457
Precision	0.9840	0.8134
Recall	0.9996	0.8989
F1-Score	0.9918	0.8541

Analysis

SVM achieves the highest training accuracy (99.17%) among all models but experiences the largest generalization gap, with test accuracy dropping to 84.57%. The near-perfect training recall (99.96%) combined with lower test precision (81.34%) suggests the model is aggressively classifying reviews as positive, leading to more false positives on unseen data.

7 Logistic Regression

Logistic Regression models the probability of class membership using a logistic function. It provides probabilistic outputs and tends to be more robust against overfitting compared to other linear methods.

Results

Metric	Training Set	Testing Set
Accuracy	0.9843	0.8678
Precision	0.9719	0.8377
Recall	0.9973	0.9139
F1-Score	0.9845	0.8742

Analysis

Logistic Regression achieves the best test performance among all models with 86.78% accuracy and the highest F1-score (87.42%). While it still shows some overfitting (training accuracy 98.43%), the gap is smaller compared to Perceptron and SVM. The model maintains strong recall (91.39%) while achieving reasonable precision (83.77%), indicating a good balance between identifying positive reviews and avoiding false positives.

8 Multinomial Naive Bayes

Multinomial Naive Bayes is based on Bayes' theorem with the assumption of feature independence. Despite this simplified assumption, it often performs surprisingly well on text classification tasks and is computationally efficient.

Results

Metric	Training Set	Testing Set
Accuracy	0.9453	0.8387
Precision	0.9722	0.8396
Recall	0.9166	0.8392
F1-Score	0.9436	0.8394

Analysis

Naive Bayes shows the smallest gap between training (94.53%) and test (83.87%) accuracy, demonstrating better generalization compared to other models. This is characteristic of Naive Bayes, which tends to be less prone to overfitting due to its strong independence assumptions. The model achieves well-balanced precision and recall on the test set (both approximately 84%), resulting in consistent performance across both classes.

Model	Test Accuracy	Test F1-Score	Overfitting Gap
Perceptron	0.8572	0.8614	13.22%
SVM	0.8457	0.8541	14.60%
Logistic Regression	**0.8678**	**0.8742**	11.65%
Naive Bayes	0.8387	0.8394	**10.66%**

9 Summary and Comparison

Key Observations:

1. **Best Overall Performance**: Logistic Regression achieved the highest test accuracy (86.78%) and F1-score (87.42%), making it the most suitable model for this sentiment analysis task.
2. **Overfitting**: All models exhibit some degree of overfitting, with training accuracies significantly higher than test accuracies. This is common in text classification with high-dimensional sparse features.
3. **Generalization**: Naive Bayes shows the best generalization with the smallest training-test gap, though its absolute performance is lower than Logistic Regression.
4. **Trade-offs**: SVM and Perceptron prioritize recall over precision, while Naive Bayes maintains a better balance between the two metrics.

The results demonstrate that for sentiment analysis on product reviews using bigram features, Logistic Regression provides the best balance of performance and generalization, making it the recommended choice for deployment.

Python Version: 3.x

All experiments used random_state=42 for reproducibility.

In [ ]:
```
# Install required libraries
!pip install pandas numpy nltk beautifulsoup4 scikit-learn matplotlib seabor
```

In [2]:
```
import pandas as pd
import numpy as np
import nltk
nltk.download('wordnet')
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
import re
from bs4 import BeautifulSoup
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.linear_model import Perceptron, LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, precision_score, recall_score, f
import warnings
warnings.filterwarnings('ignore')
```

```
[nltk_data] Downloading package wordnet to /Users/sohail/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     /Users/sohail/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt to /Users/sohail/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /Users/sohail/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
```

In [3]:
```
# Dataset path
DATA_PATH = "amazon_reviews_us_Office_Products_v1_00.tsv"
```

Dataset Preparation

Read Data

In [4]:
```
# Read the TSV file
df = pd.read_csv(DATA_PATH, sep='\t', on_bad_lines='skip')

# Display basic info
print(f"Total reviews loaded: {len(df)}")
print(f"\nColumns in dataset: {df.columns.tolist()}")
```

```
Total reviews loaded: 2640254

Columns in dataset: ['marketplace', 'customer_id', 'review_id', 'product_i
d', 'product_parent', 'product_title', 'product_category', 'star_rating', 'h
elpful_votes', 'total_votes', 'vine', 'verified_purchase', 'review_headlin
e', 'review_body', 'review_date']
```

Keep Reviews and Ratings

In [5]:

```
# Keep only review_body and star_rating columns
df = df[['review_body', 'star_rating']].copy()

# Drop rows with missing values
df = df.dropna()

# Convert star_rating to numeric (in case it's not)
df['star_rating'] = pd.to_numeric(df['star_rating'], errors='coerce')
df = df.dropna()
df['star_rating'] = df['star_rating'].astype(int)

# Display three sample reviews with ratings
print("Three Sample Reviews with Ratings:")
print("="*60)
for i, (idx, row) in enumerate(df.head(3).iterrows()):
    print(f"\nReview {i+1}:")
    print(f"Rating: {row['star_rating']} stars")
    print(f"Review: {row['review_body'][:200]}...")
    print("-"*60)

# Statistics of ratings
print("\n\nRating Statistics:")
print("="*60)
rating_counts = df['star_rating'].value_counts().sort_index()
for rating, count in rating_counts.items():
    print(f"{int(rating)}-star reviews: {count}")
```

```
Three Sample Reviews with Ratings:
============================================================

Review 1:
Rating: 5 stars
Review: Great product....
------------------------------------------------------------

Review 2:
Rating: 5 stars
Review: What's to say about this commodity item except, what have we come to
in this world.<br />Having the need to bnuy captured and compressed air. &#6
0;lol&#62;...
------------------------------------------------------------

Review 3:
Rating: 5 stars
Review: Haven't used yet, but I am sure I will like it....
------------------------------------------------------------


Rating Statistics:
============================================================
1-star reviews: 306967
2-star reviews: 138381
3-star reviews: 193680
4-star reviews: 418348
5-star reviews: 1582704
```

Relabeling and Sampling

First form three classes and print their statistics. Then randomly select 100,000 reviews from the positive and 100,000 reviews from the negative

In [6]:
```
# Create three classes based on ratings
# Positive: rating > 3 (4, 5)
# Negative: rating <= 2 (1, 2)
# Neutral: rating == 3 (will be discarded)

positive_df = df[df['star_rating'] > 3].copy()
negative_df = df[df['star_rating'] <= 2].copy()
neutral_df = df[df['star_rating'] == 3].copy()

# Print statistics for three classes
print("Class Statistics:")
print(f"Positive reviews: {len(positive_df)}")
print(f"Negative reviews: {len(negative_df)}")
print(f"Neutral reviews: {len(neutral_df)}")

# Sample 100,000 from each class (positive and negative)
positive_sample = positive_df.sample(n=100000, random_state=42)
negative_sample = negative_df.sample(n=100000, random_state=42)

# Assign binary labels
```

```
positive_sample['label'] = 1
negative_sample['label'] = 0

# Combine the samples
df_balanced = pd.concat([positive_sample, negative_sample], ignore_index=Tru

# Shuffle the combined dataset
df_balanced = df_balanced.sample(frac=1, random_state=42).reset_index(drop=T

print(f"\nBalanced dataset size: {len(df_balanced)}")
print(f"Positive samples: {len(df_balanced[df_balanced['label'] == 1])}")
print(f"Negative samples: {len(df_balanced[df_balanced['label'] == 0])}")
```

```
Class Statistics:
Positive reviews: 2001052
Negative reviews: 445348
Neutral reviews: 193680

Balanced dataset size: 200000
Positive samples: 100000
Negative samples: 100000
```

Data Cleaning

In [7]:

```
# Contractions dictionary
contractions_dict = {
    "won't": "will not", "can't": "cannot", "n't": " not", "'re": " are",
    "'s": " is", "'d": " would", "'ll": " will", "'ve": " have", "'m": " am"
    "let's": "let us", "it's": "it is", "i'm": "i am", "you're": "you are",
    "he's": "he is", "she's": "she is", "we're": "we are", "they're": "they
    "i've": "i have", "you've": "you have", "we've": "we have", "they've": "
    "i'd": "i would", "you'd": "you would", "he'd": "he would", "she'd": "sh
    "we'd": "we would", "they'd": "they would", "i'll": "i will", "you'll":
    "he'll": "he will", "she'll": "she will", "we'll": "we will", "they'll":
    "isn't": "is not", "aren't": "are not", "wasn't": "was not", "weren't":
    "haven't": "have not", "hasn't": "has not", "hadn't": "had not",
    "doesn't": "does not", "don't": "do not", "didn't": "did not",
    "wouldn't": "would not", "shouldn't": "should not", "couldn't": "could r
    "mightn't": "might not", "mustn't": "must not", "needn't": "need not",
    "shan't": "shall not", "wasn't": "was not", "weren't": "were not",
    "what's": "what is", "who's": "who is", "where's": "where is",
    "when's": "when is", "why's": "why is", "how's": "how is",
    "that's": "that is", "there's": "there is", "here's": "here is",
    "what'll": "what will", "who'll": "who will", "where'll": "where will",
    "what've": "what have", "who've": "who have", "could've": "could have",
    "would've": "would have", "should've": "should have", "might've": "might
    "must've": "must have", "y'all": "you all", "ain't": "is not",
    "gonna": "going to", "gotta": "got to", "wanna": "want to",
    "kinda": "kind of", "sorta": "sort of", "lemme": "let me", "gimme": "giv
}

# Compile contractions pattern
contractions_pattern = re.compile('({})'.format('|'.join(contractions_dict.k
                                  flags=re.IGNORECASE)
```

```
def expand_contractions(text):
    def replace(match):
        return contractions_dict.get(match.group(0).lower(), match.group(0))
    return contractions_pattern.sub(replace, text)

def clean_text(text):
    # Handle non-string values
    if not isinstance(text, str):
        return ""

    # Convert to lowercase
    text = text.lower()

    # Remove HTML tags using BeautifulSoup
    text = BeautifulSoup(text, "html.parser").get_text()

    # Remove URLs
    text = re.sub(r'http\S+|www\S+|https\S+', '', text, flags=re.MULTILINE)

    # Expand contractions
    text = expand_contractions(text)

    # Remove non-alphabetical characters (keep spaces)
    text = re.sub(r'[^a-zA-Z\s]', ' ', text)

    # Remove extra spaces
    text = re.sub(r'\s+', ' ', text).strip()

    return text

# Calculate average length before cleaning
avg_length_before_cleaning = df_balanced['review_body'].astype(str).apply(le
print(f"Average length before cleaning: {avg_length_before_cleaning:.4f}")

# Apply cleaning
df_balanced['cleaned_review'] = df_balanced['review_body'].apply(clean_text)

# Calculate average length after cleaning
avg_length_after_cleaning = df_balanced['cleaned_review'].apply(len).mean()
print(f"Average length after cleaning: {avg_length_after_cleaning:.4f}")

# Show sample before and after
print("\n\nSample Reviews - Before and After Cleaning:")
print("="*60)
for i in range(3):
    print(f"\nSample {i+1}:")
    print(f"BEFORE: {df_balanced['review_body'].iloc[i][:150]}...")
    print(f"AFTER:  {df_balanced['cleaned_review'].iloc[i][:150]}...")
    print("-"*60)
```

```
Average length before cleaning: 318.0072
Average length after cleaning: 302.9584


Sample Reviews - Before and After Cleaning:
============================================================

Sample 1:
BEFORE: This ink did not work on my Epson WF-2630.  Printer would not recogn
ize and therefore would not print.  Tried various times, rebooted printer et
c...Th...
AFTER:  this ink did not work on my epson wf printer would not recognize and
therefore would not print tried various times rebooted printer etc the custo
mer s...
------------------------------------------------------------

Sample 2:
BEFORE: These chalk markers have a great medium sized tip and erase well wit
h a wet cloth.  So far, I'm loving them and I hope they last throughout the
school...
AFTER:  these chalk markers have a great medium sized tip and erase well wit
h a wet cloth so far i am loving them and i hope they last throughout the sc
hool y...
------------------------------------------------------------

Sample 3:
BEFORE: The ink runs out too fast. Not very good product....
AFTER:  the ink runs out too fast not very good product...
------------------------------------------------------------
```

Pre-processing

remove the stop words

In [8]:
```
from nltk.corpus import stopwords

# Get English stopwords
stop_words = set(stopwords.words('english'))

def remove_stopwords(text):
    words = text.split()
    filtered_words = [word for word in words if word.lower() not in stop_wor
    return ' '.join(filtered_words)

# Store cleaned review before preprocessing for comparison
df_balanced['before_preprocessing'] = df_balanced['cleaned_review'].copy()
```

perform lemmatization

In [9]:
```
from nltk.stem import WordNetLemmatizer
```

```
lemmatizer = WordNetLemmatizer()

def lemmatize_text(text):
    words = text.split()
    lemmatized_words = [lemmatizer.lemmatize(word) for word in words]
    return ' '.join(lemmatized_words)

def preprocess_text(text):
    # Remove stopwords
    text = remove_stopwords(text)
    # Lemmatize
    text = lemmatize_text(text)
    return text

# Calculate average length before preprocessing
avg_length_before_preprocessing = df_balanced['cleaned_review'].apply(len).m
print(f"Average length before preprocessing: {avg_length_before_preprocessir

# Apply preprocessing
df_balanced['preprocessed_review'] = df_balanced['cleaned_review'].apply(pre

# Calculate average length after preprocessing
avg_length_after_preprocessing = df_balanced['preprocessed_review'].apply(le
print(f"Average length after preprocessing: {avg_length_after_preprocessing:

# Show three sample reviews before and after data cleaning + preprocessing
print("\n\nThree Sample Reviews - Before and After Data Cleaning + Preproces
print("="*80)
for i in range(3):
    print(f"\nSample {i+1}:")
    print(f"ORIGINAL:      {df_balanced['review_body'].iloc[i][:100]}...")
    print(f"AFTER CLEAN:   {df_balanced['cleaned_review'].iloc[i][:100]}..."
    print(f"AFTER PREPROC: {df_balanced['preprocessed_review'].iloc[i][:100]
    print("-"*80)
```

```
Average length before preprocessing: 302.9584
Average length after preprocessing: 187.4218


Three Sample Reviews - Before and After Data Cleaning + Preprocessing:
==========================================================================
====

Sample 1:
ORIGINAL:      This ink did not work on my Epson WF-2630.  Printer would not
recognize and therefore would not prin...
AFTER CLEAN:   this ink did not work on my epson wf printer would not recogn
ize and therefore would not print tried...
AFTER PREPROC: ink work epson wf printer would recognize therefore would pri
nt tried various time rebooted printer ...
--------------------------------------------------------------------------
----

Sample 2:
ORIGINAL:      These chalk markers have a great medium sized tip and erase w
ell with a wet cloth.  So far, I'm lovi...
AFTER CLEAN:   these chalk markers have a great medium sized tip and erase w
ell with a wet cloth so far i am loving...
AFTER PREPROC: chalk marker great medium sized tip erase well wet cloth far
loving hope last throughout school year...
--------------------------------------------------------------------------
----

Sample 3:
ORIGINAL:      The ink runs out too fast. Not very good product....
AFTER CLEAN:   the ink runs out too fast not very good product...
AFTER PREPROC: ink run fast good product...
--------------------------------------------------------------------------
----
```

Bigram Feature Extraction

In [10]:

```python
# Extract Bigram features using CountVectorizer
# ngram_range=(2,2) means only bigrams

vectorizer = CountVectorizer(ngram_range=(2, 2))

# Fit and transform the preprocessed reviews
X = vectorizer.fit_transform(df_balanced['preprocessed_review'])
y = df_balanced['label'].values

print(f"Feature matrix shape: {X.shape}")
print(f"Number of bigram features: {X.shape[1]}")

# Split into training and testing sets (80/20)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
print(f"\nTraining set size: {X_train.shape[0]}")
print(f"Testing set size: {X_test.shape[0]}")
```

```
Feature matrix shape: (200000, 1836589)
Number of bigram features: 1836589

Training set size: 160000
Testing set size: 40000
```

Perceptron

In [11]:

```
# Helper function to print metrics
def print_metrics(model_name, y_true, y_pred, split_name):
    accuracy = accuracy_score(y_true, y_pred)
    precision = precision_score(y_true, y_pred)
    recall = recall_score(y_true, y_pred)
    f1 = f1_score(y_true, y_pred)

    print(f"{model_name} {split_name} Accuracy: {accuracy:.4f}")
    print(f"{model_name} {split_name} Precision: {precision:.4f}")
    print(f"{model_name} {split_name} Recall: {recall:.4f}")
    print(f"{model_name} {split_name} F1-score: {f1:.4f}")

    return accuracy, precision, recall, f1

# Train Perceptron model
perceptron = Perceptron(random_state=42)
perceptron.fit(X_train, y_train)

# Predictions
y_train_pred_perceptron = perceptron.predict(X_train)
y_test_pred_perceptron = perceptron.predict(X_test)

# Print metrics
print("="*50)
print("PERCEPTRON RESULTS")
print("="*50)
print_metrics("Perceptron", y_train, y_train_pred_perceptron, "Training")
print()
print_metrics("Perceptron", y_test, y_test_pred_perceptron, "Testing")
```

```
==================================================
PERCEPTRON RESULTS
==================================================
Perceptron Training Accuracy: 0.9894
Perceptron Training Precision: 0.9833
Perceptron Training Recall: 0.9958
Perceptron Training F1-score: 0.9895

Perceptron Testing Accuracy: 0.8572
Perceptron Testing Precision: 0.8409
Perceptron Testing Recall: 0.8829
Perceptron Testing F1-score: 0.8614
```

Out[11]:

```
(0.85725, 0.8409327898378993, 0.8828622611464968, 0.8613875807156381)
```

SVM

In [12]:
```
# Train SVM model (using LinearSVC for efficiency)
svm = LinearSVC(random_state=42)
svm.fit(X_train, y_train)

# Predictions
y_train_pred_svm = svm.predict(X_train)
y_test_pred_svm = svm.predict(X_test)

# Print metrics
print("="*50)
print("SVM RESULTS")
print("="*50)
print_metrics("SVM", y_train, y_train_pred_svm, "Training")
print()
print_metrics("SVM", y_test, y_test_pred_svm, "Testing")
```

```
==================================================
SVM RESULTS
==================================================
SVM Training Accuracy: 0.9917
SVM Training Precision: 0.9840
SVM Training Recall: 0.9996
SVM Training F1-score: 0.9918

SVM Testing Accuracy: 0.8457
SVM Testing Precision: 0.8134
SVM Testing Recall: 0.8989
SVM Testing F1-score: 0.8541
```

Out[12]:
```
(0.84565, 0.8134456051873199, 0.8989351114649682, 0.8540563540090772)
```

Logistic Regression

In [13]:
```
# Train Logistic Regression model
lr = LogisticRegression(random_state=42, max_iter=1000)
lr.fit(X_train, y_train)

# Predictions
y_train_pred_lr = lr.predict(X_train)
y_test_pred_lr = lr.predict(X_test)

# Print metrics
print("="*50)
print("LOGISTIC REGRESSION RESULTS")
print("="*50)
print_metrics("Logistic Regression", y_train, y_train_pred_lr, "Training")
print()
print_metrics("Logistic Regression", y_test, y_test_pred_lr, "Testing")
```

```
==================================================
LOGISTIC REGRESSION RESULTS
==================================================
Logistic Regression Training Accuracy: 0.9843
Logistic Regression Training Precision: 0.9719
Logistic Regression Training Recall: 0.9973
Logistic Regression Training F1-score: 0.9845

Logistic Regression Testing Accuracy: 0.8678
Logistic Regression Testing Precision: 0.8377
Logistic Regression Testing Recall: 0.9139
Logistic Regression Testing F1-score: 0.8742
```

Out[13]: (0.8678, 0.837712096332786, 0.9139132165605095, 0.8741551642075203)

Naive Bayes

In [14]:
```
# Train Multinomial Naive Bayes model
nb = MultinomialNB()
nb.fit(X_train, y_train)

# Predictions
y_train_pred_nb = nb.predict(X_train)
y_test_pred_nb = nb.predict(X_test)

# Print metrics
print("="*50)
print("NAIVE BAYES RESULTS")
print("="*50)
print_metrics("Naive Bayes", y_train, y_train_pred_nb, "Training")
print()
print_metrics("Naive Bayes", y_test, y_test_pred_nb, "Testing")
```

```
==================================================
NAIVE BAYES RESULTS
==================================================
Naive Bayes Training Accuracy: 0.9453
Naive Bayes Training Precision: 0.9722
Naive Bayes Training Recall: 0.9166
Naive Bayes Training F1-score: 0.9436

Naive Bayes Testing Accuracy: 0.8387
Naive Bayes Testing Precision: 0.8396
Naive Bayes Testing Recall: 0.8392
Naive Bayes Testing F1-score: 0.8394
```

Out[14]: (0.83865, 0.8395897640147366, 0.839171974522293, 0.8393808172813698)

Advanced Analysis and Visualizations

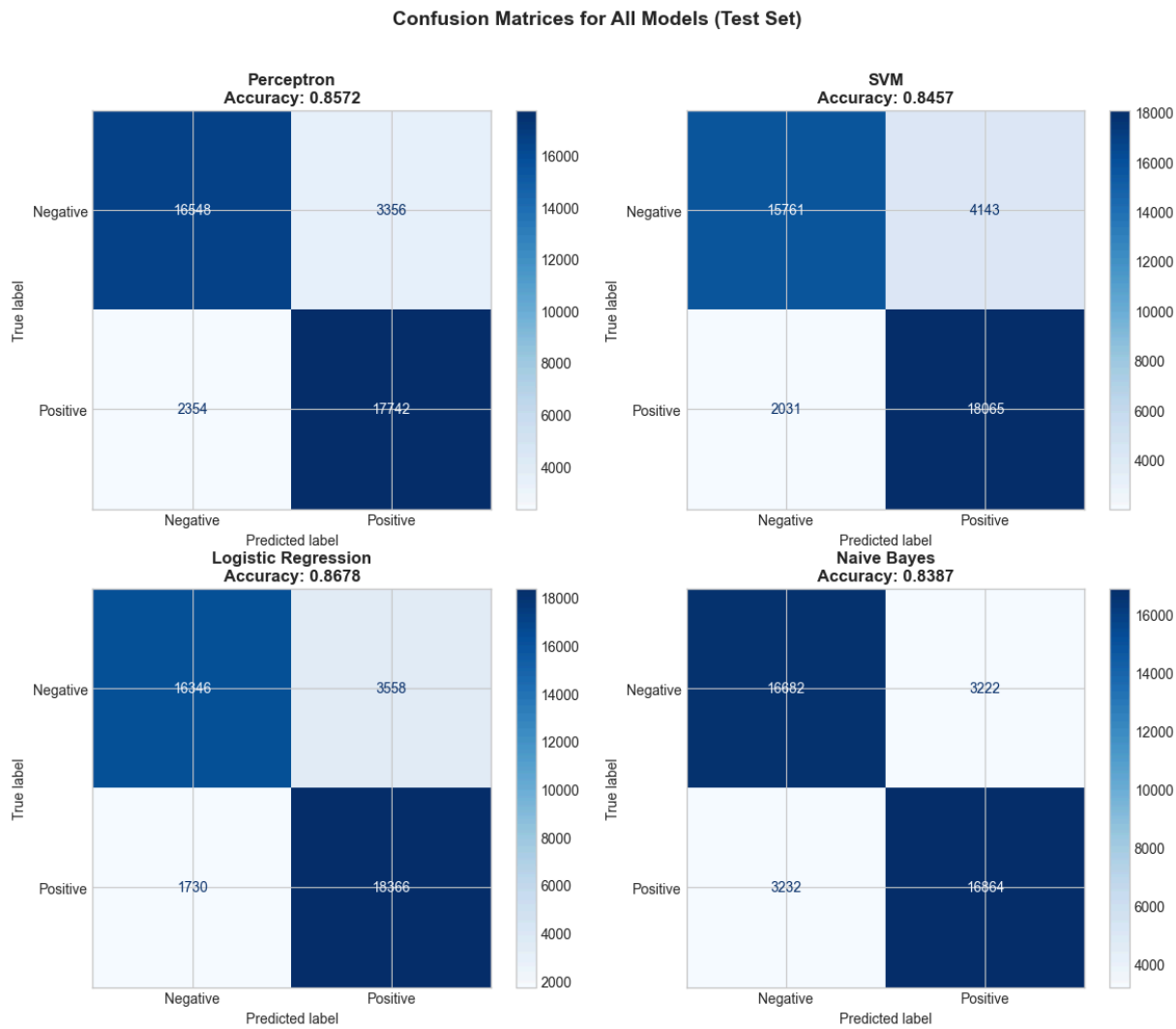
The following sections provide deeper insights into model performance through visualizations and error analysis.

In [15]:
```
# Import visualization libraries
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

# Set style for better looking plots
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("husl")
```

Confusion Matrices

Confusion matrices provide a detailed view of model predictions, showing true positives, true negatives, false positives, and false negatives.

In [16]:
```
# Create confusion matrices for all models
fig, axes = plt.subplots(2, 2, figsize=(12, 10))

models_data = [
    ('Perceptron', y_test_pred_perceptron),
    ('SVM', y_test_pred_svm),
    ('Logistic Regression', y_test_pred_lr),
    ('Naive Bayes', y_test_pred_nb)
]

for ax, (name, y_pred) in zip(axes.flat, models_data):
    cm = confusion_matrix(y_test, y_pred)
    disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=['Nega
    disp.plot(ax=ax, cmap='Blues', values_format='d')
    ax.set_title(f'{name}\nAccuracy: {accuracy_score(y_test, y_pred):.4f}',

plt.suptitle('Confusion Matrices for All Models (Test Set)', fontsize=14, fo
plt.tight_layout()
plt.savefig('confusion_matrices.png', dpi=150, bbox_inches='tight')
plt.show()
```

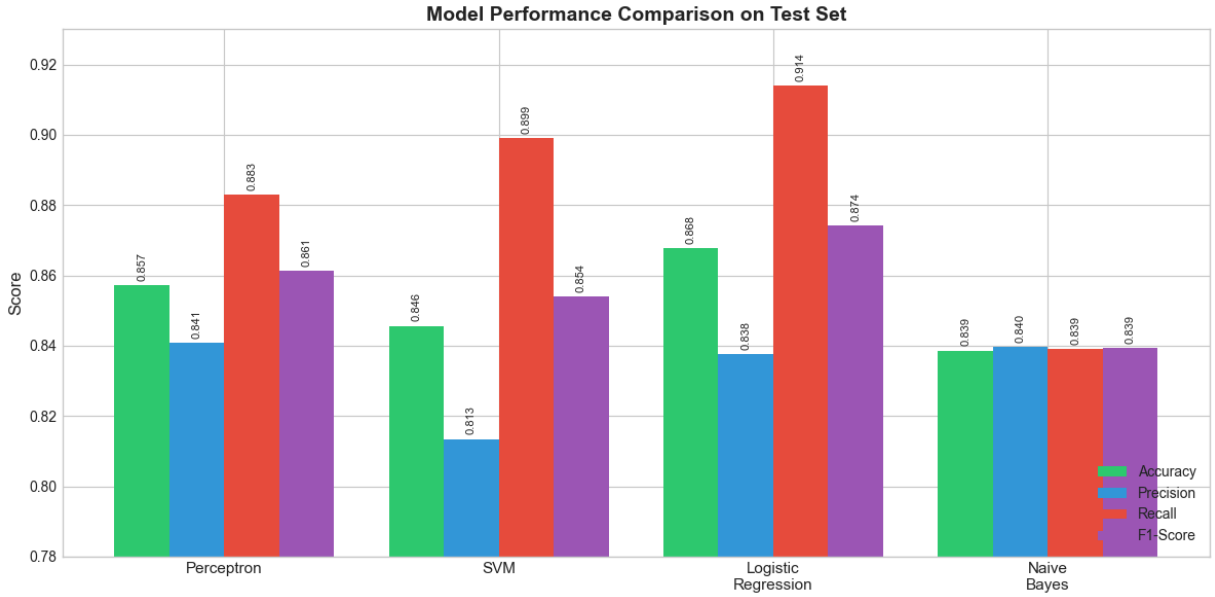


Model Performance Comparison

A visual comparison of all models across key metrics helps identify the best performer at a glance.

In [17]:
```
# Model comparison visualization
models = ['Perceptron', 'SVM', 'Logistic\nRegression', 'Naive\nBayes']

# Calculate metrics for each model
metrics_data = {
    'Accuracy': [
        accuracy_score(y_test, y_test_pred_perceptron),
        accuracy_score(y_test, y_test_pred_svm),
        accuracy_score(y_test, y_test_pred_lr),
        accuracy_score(y_test, y_test_pred_nb)
    ],
    'Precision': [
        precision_score(y_test, y_test_pred_perceptron),
        precision_score(y_test, y_test_pred_svm),
        precision_score(y_test, y_test_pred_lr),
        precision_score(y_test, y_test_pred_nb)
```

```
    ],
    'Recall': [
        recall_score(y_test, y_test_pred_perceptron),
        recall_score(y_test, y_test_pred_svm),
        recall_score(y_test, y_test_pred_lr),
        recall_score(y_test, y_test_pred_nb)
    ],
    'F1-Score': [
        f1_score(y_test, y_test_pred_perceptron),
        f1_score(y_test, y_test_pred_svm),
        f1_score(y_test, y_test_pred_lr),
        f1_score(y_test, y_test_pred_nb)
    ]
}

x = np.arange(len(models))
width = 0.2

fig, ax = plt.subplots(figsize=(12, 6))

bars1 = ax.bar(x - 1.5*width, metrics_data['Accuracy'], width, label='Accura
bars2 = ax.bar(x - 0.5*width, metrics_data['Precision'], width, label='Preci
bars3 = ax.bar(x + 0.5*width, metrics_data['Recall'], width, label='Recall',
bars4 = ax.bar(x + 1.5*width, metrics_data['F1-Score'], width, label='F1-Sco

ax.set_ylabel('Score', fontsize=12)
ax.set_title('Model Performance Comparison on Test Set', fontsize=14, fontwe
ax.set_xticks(x)
ax.set_xticklabels(models, fontsize=11)
ax.legend(loc='lower right', fontsize=10)
ax.set_ylim(0.78, 0.93)

# Add value labels on bars
for bars in [bars1, bars2, bars3, bars4]:
    for bar in bars:
        height = bar.get_height()
        ax.annotate(f'{height:.3f}',
                    xy=(bar.get_x() + bar.get_width() / 2, height),
                    xytext=(0, 3),
                    textcoords="offset points",
                    ha='center', va='bottom', fontsize=8, rotation=90)

plt.tight_layout()
plt.savefig('model_comparison.png', dpi=150, bbox_inches='tight')
plt.show()

# Print best model
best_idx = np.argmax(metrics_data['F1-Score'])
print(f"\nBest Model: {models[best_idx].replace(chr(10), ' ')} with F1-Score
```




```
Best Model: Logistic Regression with F1-Score: 0.8742
```

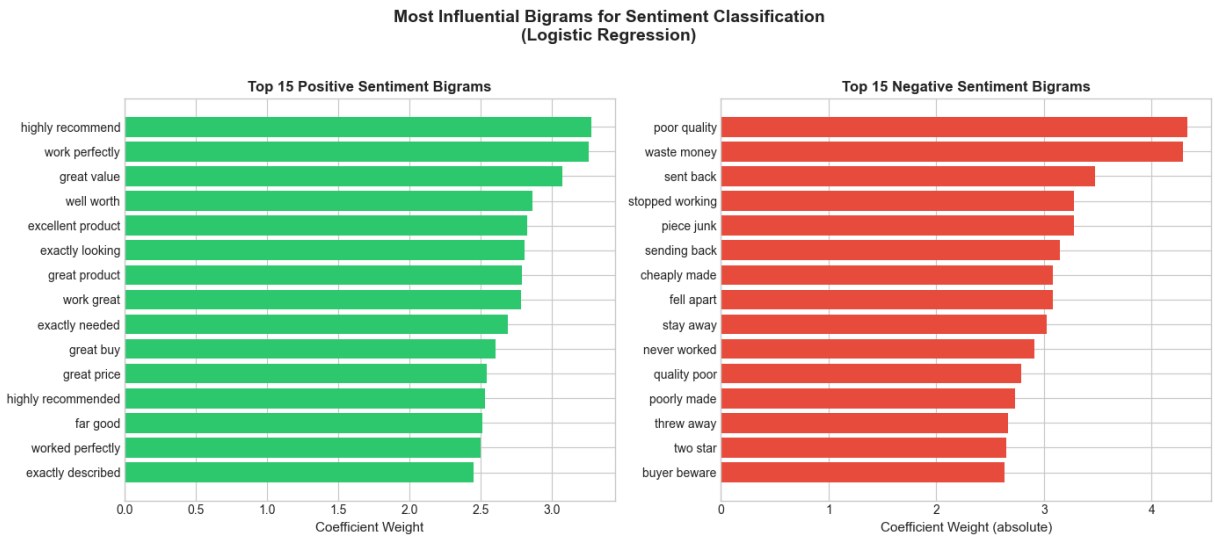
Top Bigrams Analysis

Analyzing the most influential bigrams for positive and negative sentiment provides interpretable insights into what the model has learned.

In [18]:

```
# Extract top bigrams from Logistic Regression (best model)
feature_names = vectorizer.get_feature_names_out()
coefs = lr.coef_[0]

# Get indices of top positive and negative bigrams
top_positive_idx = coefs.argsort()[-15:][::-1]
top_negative_idx = coefs.argsort()[:15]

top_positive_bigrams = [(feature_names[i], coefs[i]) for i in top_positive_i
top_negative_bigrams = [(feature_names[i], coefs[i]) for i in top_negative_i

# Create visualization
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Positive bigrams
pos_bigrams = [b[0] for b in top_positive_bigrams]
pos_weights = [b[1] for b in top_positive_bigrams]
axes[0].barh(pos_bigrams, pos_weights, color='#2ecc71')
axes[0].set_xlabel('Coefficient Weight', fontsize=11)
axes[0].set_title('Top 15 Positive Sentiment Bigrams', fontsize=12, fontweig
axes[0].invert_yaxis()

# Negative bigrams
neg_bigrams = [b[0] for b in top_negative_bigrams]
neg_weights = [abs(b[1]) for b in top_negative_bigrams]
axes[1].barh(neg_bigrams, neg_weights, color='#e74c3c')
axes[1].set_xlabel('Coefficient Weight (absolute)', fontsize=11)
axes[1].set_title('Top 15 Negative Sentiment Bigrams', fontsize=12, fontweig
axes[1].invert_yaxis()
```

```
plt.suptitle('Most Influential Bigrams for Sentiment Classification\n(Logist
             fontsize=14, fontweight='bold', y=1.02)
plt.tight_layout()
plt.savefig('top_bigrams.png', dpi=150, bbox_inches='tight')
plt.show()

# Print the bigrams
print("\nTop 15 Positive Sentiment Bigrams:")
for bigram, weight in top_positive_bigrams:
    print(f"  '{bigram}': {weight:.4f}")

print("\nTop 15 Negative Sentiment Bigrams:")
for bigram, weight in top_negative_bigrams:
    print(f"  '{bigram}': {weight:.4f}")
```



```
Top 15 Positive Sentiment Bigrams:
  'highly recommend': 3.2758
  'work perfectly': 3.2550
  'great value': 3.0749
  'well worth': 2.8632
  'excellent product': 2.8235
  'exactly looking': 2.8074
  'great product': 2.7900
  'work great': 2.7797
  'exactly needed': 2.6919
  'great buy': 2.6017
  'great price': 2.5411
  'highly recommended': 2.5265
  'far good': 2.5123
  'worked perfectly': 2.4980
  'exactly described': 2.4503

Top 15 Negative Sentiment Bigrams:
  'poor quality': -4.3359
  'waste money': -4.2892
  'sent back': -3.4750
  'stopped working': -3.2821
  'piece junk': -3.2811
  'sending back': -3.1518
  'cheaply made': -3.0851
  'fell apart': -3.0818
  'stay away': -3.0237
  'never worked': -2.9139
  'quality poor': -2.7929
  'poorly made': -2.7320
  'threw away': -2.6629
  'two star': -2.6542
  'buyer beware': -2.6362
```

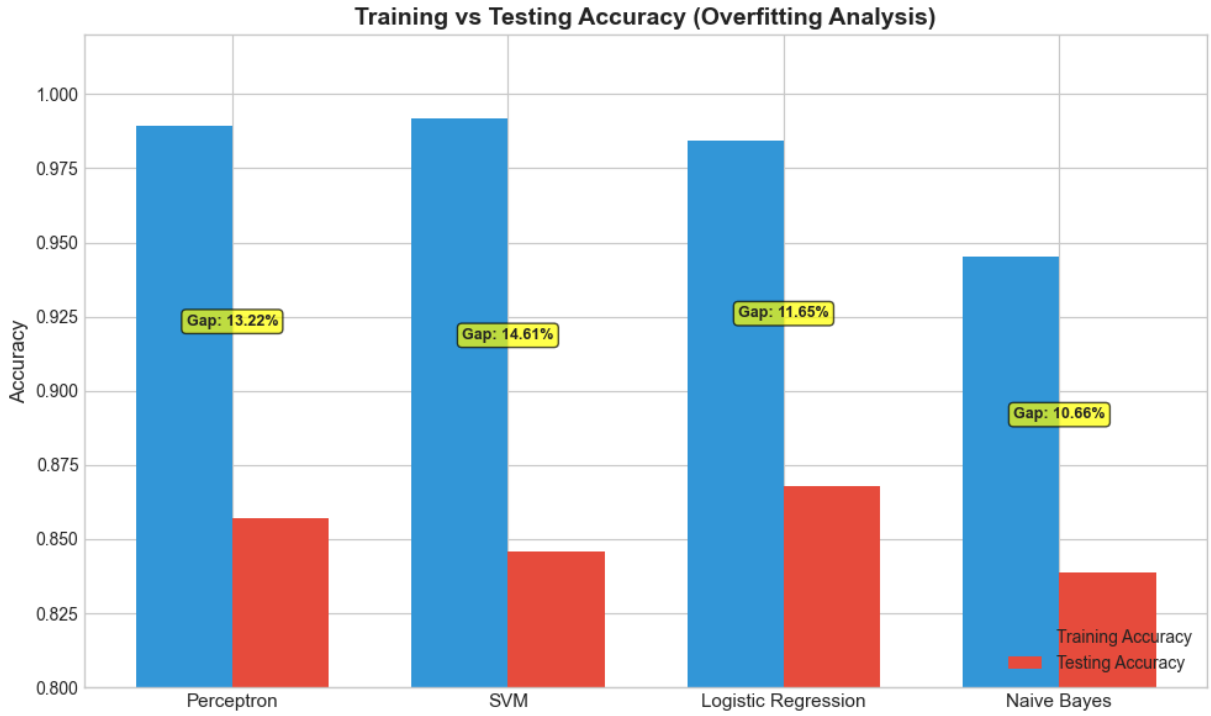
Overfitting Analysis

Comparing training vs testing performance reveals the generalization capability of each model.

In [19]:

```
# Overfitting Analysis - Training vs Testing Accuracy
models_names = ['Perceptron', 'SVM', 'Logistic Regression', 'Naive Bayes']

train_accuracies = [
    accuracy_score(y_train, y_train_pred_perceptron),
    accuracy_score(y_train, y_train_pred_svm),
    accuracy_score(y_train, y_train_pred_lr),
    accuracy_score(y_train, y_train_pred_nb)
]

test_accuracies = [
    accuracy_score(y_test, y_test_pred_perceptron),
    accuracy_score(y_test, y_test_pred_svm),
    accuracy_score(y_test, y_test_pred_lr),
    accuracy_score(y_test, y_test_pred_nb)
```

```
]

overfitting_gap = [train - test for train, test in zip(train_accuracies, tes

# Create visualization
fig, ax = plt.subplots(figsize=(10, 6))

x = np.arange(len(models_names))
width = 0.35

bars1 = ax.bar(x - width/2, train_accuracies, width, label='Training Accurac
bars2 = ax.bar(x + width/2, test_accuracies, width, label='Testing Accuracy'

ax.set_ylabel('Accuracy', fontsize=12)
ax.set_title('Training vs Testing Accuracy (Overfitting Analysis)', fontsize
ax.set_xticks(x)
ax.set_xticklabels(models_names, fontsize=11)
ax.legend(loc='lower right', fontsize=10)
ax.set_ylim(0.8, 1.02)

# Add gap annotations
for i, (train, test, gap) in enumerate(zip(train_accuracies, test_accuracies
    ax.annotate(f'Gap: {gap:.2%}',
                xy=(i, (train + test) / 2),
                ha='center', va='center',
                fontsize=9, fontweight='bold',
                bbox=dict(boxstyle='round', facecolor='yellow', alpha=0.7))

plt.tight_layout()
plt.savefig('overfitting_analysis.png', dpi=150, bbox_inches='tight')
plt.show()

# Summary table
print("\nOverfitting Summary:")
print("-" * 60)
print(f"{'Model':<22} {'Train Acc':<12} {'Test Acc':<12} {'Gap':<12}")
print("-" * 60)
for name, train, test, gap in zip(models_names, train_accuracies, test_accur
    print(f"{name:<22} {train:<12.4f} {test:<12.4f} {gap:<12.2%}")
print("-" * 60)
print(f"\nNaive Bayes shows the least overfitting (best generalization)")
print(f"Logistic Regression achieves the best test accuracy despite moderate
```




```
Overfitting Summary:
------------------------------------------------------------
Model                  Train Acc    Test Acc     Gap
------------------------------------------------------------
Perceptron             0.9894       0.8572       13.22%
SVM                    0.9917       0.8457       14.61%
Logistic Regression    0.9843       0.8678       11.65%
Naive Bayes            0.9453       0.8387       10.66%
------------------------------------------------------------

Naive Bayes shows the least overfitting (best generalization)
Logistic Regression achieves the best test accuracy despite moderate overfit
ting
```